SEHS4654 Capstone Project

BRIEFING

24-AUG-2025

Contacts

1. Your supervisor
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   - For project coordination

Intended Learning Outcomes

Upon completion of the subject, students will be able:

(a) apply specialised knowledge independently and as a team;

(b) communicate progress and achievement orally and in written reports;

(c) formulate a creative, rich in intellectual content and challenging project to solve key engineering problems;

(d) examine the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions;

(e) synthesize relevant knowledge and analytical skills gained in various engineering domains;

(f) develop professionalism with self-confidence and independence by successfully completing the project in a competent manner.

Deliverables

Tasks	%	Due Date
Assessment of student's performance	10%	-
Formal project proposal	5%	Week 4 – Sem 1
Interim progress report	20%	Week 19 – Sem 1
Final report	40%	Week 13 – Sem 2
Presentation and demonstration	25%	Week 14 – Sem 2
Total	100%	

Month	Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sem Week
Aug 2025	-	25	26	27	28	29	30	31	--
Sep	1	1	2	3	4	5	6	7	1
	2	8	9	10	11	12	13	14	2
	3	15	16	17	18	19	20	21	3
	4	22	23	24	25	26	27	28	4
Oct	5	29	30	1	2	3	4	5	5
	6	6	7	8	9	10	11	12	6
	7	13	14	15	16	17	18	19	7
	8	20	21	22	23	24	25	26	8
Nov	9	27	28	29	30	31	1	2	9
	10	3	4	5	6	7	8	9	10
	11	10	11	12	13	14	15	16	11
	12	17	18	19	20	21	22	23	12
	13	24	25	26	27	28	29	30	13
Dec	14	1	2	3	4	5	6	7	Revision / Exam
	15	8	9	10	11	12	13	14	Exam
	16	15	16	17	18	19	20	21	Exam
	17	22	23	24	25	26	27	28	Exam
Jan 2026	18	29	30	31	1	2	3	4	[illegible]
	19	5	6	7	8	9	10	11	[illegible] result
	20	12	13	14	15	16	17	18	[illegible] processing
	21	19	20	21	22	23	24	25	1

Month	Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sem Week
	21	19	20	21	22	23	24	25	1
Feb	22	26	27	28	29	30	31	1	2
	23	2	3	4	5	6	7	8	3
	24	9	10	11	12	13	14	15	4
	25	16	17	18	19	20	21	22	Lunar New Year Break
Mar	26	23	24	25	26	27	28	1	5
	27	2	3	4	5	6	7	8	6
	28	9	10	11	12	13	14	15	7
	29	16	17	18	19	20	21	22	8
	30	23	24	25	26	27	28	29	9
Apr	31	30	31	1	2	3	4	5	10
	32	6	7	8	9	10	11	12	11
	33	13	14	15	16	17	18	19	12
	34	20	21	22	23	24	25	26	13
May	35	27	28	29	30	1	2	3	Revision / Exam

Project Types

- Research-based
  Involve lab experiments, design and development of test rigs, computer simulations, data analysis and interpretations
- Design-based
  Involve design of products/systems, computer simulations and analysis, prototype building and testing, data analysis and interpretations

Formal Project Proposal

- Outputs from preliminary stage of the capstone project, relevant literature reviews and scheme design or research plan.
  - Aims of the project
  - Proposed specifications of the product
    (no matter it is a hardware or software project)
  - Summary of the literature search did up-to-date
  - Proposed approach/methodology to be used
  - Some brief descriptions on the theory of the approach/methodology
  - Timetable of your work to the end of the project
- 8-15 pages, excluding appendix
- Submit by Week 4 – Sem 1
- Assessed by your supervisor
- Check with Turnitin

Interim Report

- Outputs from initial steps of the capstone project
- Detailed plan of activities up to the completion of the project
- Major portion of this report may go into the final report
- Assume the reader is an engineer having a basic knowledge of the subject
- The report should be clear and concise
- Approximately 20 pages of single line spacing
- Font: Times New Roman, size 12 pt.
- Check with Turnitin
- Assessed by supervisor and an independent assessor
- Submit by Week 19 – Sem 1

Interim Report Structure

- Title page
- Table of contents
- Introduction
- Aims and objectives
- Methodology
- Literature review/ market research (as appropriate)
- Others (progress made in the project activities organized under appropriate headings)
- Project management (project planning details: Gantt chart, work breakdown and brief evaluation of progress made)
- Conclusion
- References (Use APA referencing style)
- Appendix
  - Log sheets
  - Any other relevant materials as required

Final Report

- Outputs from all the activities undertaken for the capstone project
- A considerable portion of this report may come from the interim report
- Minimum 45 pages, maximum 200 pages, normally 60 pages
- Single line spacing
- Font: Times New Roman, size 12 pt.
- Check with Turnitin
- Assessed by the supervisor and an independent assessor
- Submit by Week 13 – Sem 2

Final Report Structure

- Title page
- Capstone Project Declaration Form
- Acknowledgement
- Table of contents
- Abstract
- Nomenclature
- Chapter 1: Introduction
- Chapter 2 – P: The main content of the report
- Chapter Q: Conclusions and possible recommendations for future work
- References
- Appendices
- Log sheets

Presentation and Demonstration

- Every group member presents
  - The project
  - Their contribution to the whole project
  - Respond to the questions addressed to them by the examination panel
- Assessed by examination panel
- Assessment criteria
  - Technical concept/knowledge/application
  - Intellectual level, response to questions
  - Demonstration and engineering accomplishment
  - Presentation skill and language competence
- In week 14, Sem 2

Assessment of Student's Performance

- Conducted by your supervisor
- Motivation and perseverance
- Originality and innovation of the project
- Execution and problem-solving skills
- Communication
- Self-discipline and time management

Extended Submission

- Apply through your supervisor
- Extended to end of summer term
- Maximum grade: C

Advices

- Meet you supervisor once per month, or on need basis
- Meet your groupmates at least once a week
- Work harder in the winter break
- Divide works and coordinate
- Maintain meeting log sheets
- Refer to the handbook for title page templates

Laboratory Available

- WK-S1107 Electrical Engineering and Physics Laboratory
  - Basic hand/electric tool
  - Basic electric measurement: differential volt probes, current probes
  - Three phase power
  - Soldering
  - Temporary material storage

Reimbursement

- Keep the original receipts
  - Make photo copies of thermo-print receipts, submit both copies and originals
- Keep the PDF record for online payment
- Claim after completing the project
- Maximum amount as announced in project proposal (to be approved)